

Junhwi Cho

Ph. D. Candidate, KAIST

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RESEARCH INTERESTS

- Tactile sensing
- Robotic skin
- Soft robotics
- Wearable robot

EDUCATION

Ph.D. in Mechanical Engineering Feb. 2022 – Feb. 2027 (Expected)

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea
Advisor: Professor Jung Kim

M.S. in Mechanical Engineering Feb. 2020 - Feb. 2022

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea
Advisor: Professor Jung Kim
(Thesis: A Soft Somesthetic Robotic Finger based on 3D Electrical Resistance Tomography and Hydrogel)

B.S. in Mechanical Engineering Feb. 2015 - Feb. 2020

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

PUBLICATIONS

Journals

- [J1] Junhwan Choi, Jirou Feng, **Junhwi Cho**, and Jung Kim. “Development of a Sensor Suit for Gait Environment Detection Using Non-contact Sensors and Integrated Model”. *IEEE/ASME Transactions on Mechatronics*. 2025
- [J2] Kyungseo Park, Hyunwoo Yuk, Minjin Yang, **Junhwi Cho**, Hyosang Lee, and Jung Kim. “A biomimetic elastomeric robot skin using electrical impedance and acoustic tomography for tactile sensing”. *Science Robotics*. 2022.

- [J3] Hyunkyu Park, **Junhwi Cho**, Junghoon Park, Youngjin Na, and Jung Kim. “Sim-to-real Transfer Learning Approach for Tracking Multi-DOF Ankle Motions using Soft Strain Sensors”. *IEEE Robotics and Automation Letters*. 2020..
- [J4] **Junhwi Cho**, Hyunkyu Park, and Jung Kim. “Development of Wearable Sensing Suit for Monitoring Wrist Joint Motions and Deep Neural Network-based Calibration Method”. In: *Journal of the Korean Society for Precision Engineering (JKSPE)*. 2020.

International Conferences

- [C1] **Junhwi Cho***, Sunggyu Bae*, Junghyeon Ma*, Hyosang lee, Jung Kim, and Kyungseo Park. “EIT-Pneumatic Hybrid Robotic Skin for Practical and Accurate Force Map Reconstruction”. *IEEE International Conference on Robotics & Automation (ICRA)*. 2026. (*: equal contribution)
- [C2] Minju Kim, **Junhwi Cho**, Sangha Jeon, Hyun Ki Park, Woojong Kim, and Jung Kim. “Robotic Skin with Sparsely Embedded Transducers for Contact Localization on Non-Planar Surfaces”. *International Conference on Ubiquitous Robots (UR)*, 2026.
- [C3] Hyun Ki Park, Yoonjin Kim, Sungbin Park, **Junhwi Cho**, Minju Kim, Dayoung Yeom, and Jung Kim. “Design of a Vision-Integrated Suction Cup for Drone Perching”. *International Conference on Ubiquitous Robots (UR)*, 2026.
- [C4] **Junhwi Cho**, Hyunjo Chung, Kyungseo Park and Jung Kim. “Spatial Sensitivity Equalization of ERT-based Robotic Skin through Gauge Factor Distribution Optimization”. *IEEE International Conference on Robotics & Automation (ICRA)*. 2025.
- [C5] MinJin Yang, **Junhwi Cho**, Hyunjo Chung, Kyungseo Park, and Jung Kim. “Touch Classification on Robotic Skin using Multimodal Tactile Sensing Modules”. *IEEE International Conference on Robotics & Automation (ICRA)*. 2023.
- [C6] **Junhwi Cho**, and Jung Kim. “Simulation Study of Electrical Resistance Tomography-based Tactile Sensors: Spatial Resolution and Image Performance Evaluation”, *International Conference on Ubiquitous Robots (UR)*. 2023.
- [C7] Ung Heo, Hyunho Jeong, Jirou Feng, **Junhwi Cho**, Kyungseo Park, Youngsik Uoon, Dongyeon Lee, and Jung Kim. “Gait Phase Detection using Bioimpedance and sEMG Combined Sensor”. *International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 2023.
- [C8] **Junhwi Cho**, Kyungseo Park, Hwayeong Jeong, and Jung Kim. “A Soft Somesthetic Robotic Finger Based on Conductive Working Liquid and an Origami Structure”. *IEEE/RSJ*

International Conference on Intelligent Robots and Systems (IROS). 2021.

HONORS AND AWARDS

- Silver prize, The 28th Humantech paper award, 2022
- Best Student Paper Award, Korea Robotics Society Annual Conference, 2019

RESEARCH EXPERIENCE

Venture Research Program for Master's and PhD students May. 2021 - Dec. 2021

Biorobotics Laboratory (Advisor: Prof. Jung Kim), KAIST

- Skin-Inspired Biomimetic Soft Robotic Gripper

Undergraduate Research Program (URP) Jun. 2018 - Dec. 2018

Biorobotics Laboratory (Advisor: Prof. Jung Kim), KAIST

- Development of real-time motion detection wearable suit based on machine learning for measurement accuracy improvement

ACADEMIC SERVICES

Conference Reviewer:

- IEEE International Conference on Robotics & Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

TEACHING EXPERIENCE

Teaching Assistant, Mechanical Engineering Laboratory Mar. 2022 - Jun. 2023

Dept. of Mechanical Engineering, KAIST

Teaching Assistant, Mechatronics System Design Sep. 2022 - Dec. 2022

Dept. of Mechanical Engineering, KAIST

Teaching Assistant, Automatic Control Mar. 2023 - Jun. 2023

Dept. of Mechanical Engineering, KAIST

Teaching Assistant, Freshman Seminar <Department of Mechanical Engineering> Sep. 2023 - Dec. 2024

Dept. of Mechanical Engineering, KAIST

Counselling Assistant
Dept. of Mechanical Engineering, KAIST

Sep. 2023 - Jun. 2024

Applied Electronics
Dept. of Mechanical Engineering, KAIST

Mar. 2024 - Jun. 2024